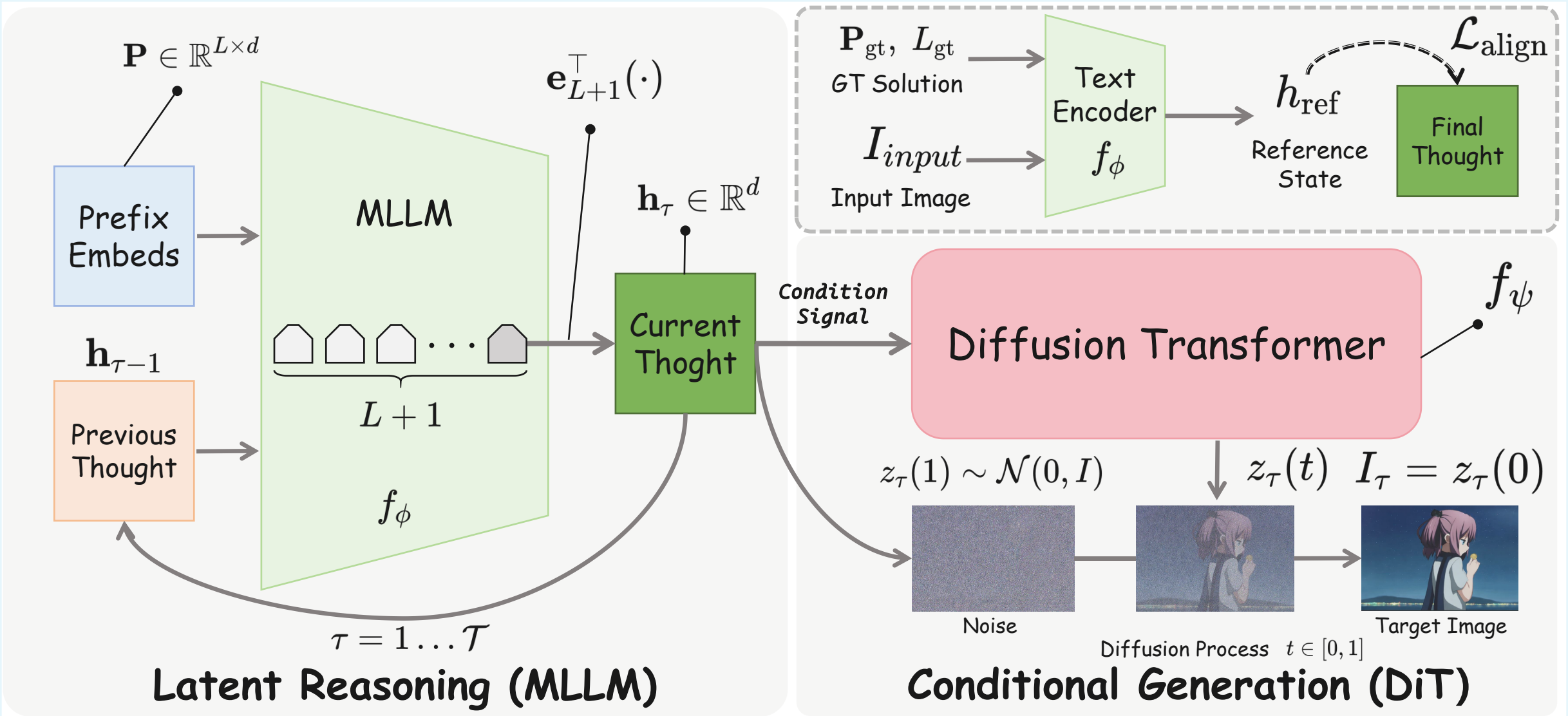
$\mathbf{P} \in \mathbb{R}^{L \times d}$
Prefix Embeds
$\mathbf{h}_{\tau-1}$
Previous Thought
MLLM
$L+1$
f_ϕ
$\mathbf{e}_{L+1}^\top(\cdot)$
$\mathbf{h}_\tau \in \mathbb{R}^d$
Current Thoght
$\tau = 1 \ldots \mathcal{T}$
Latent Reasoning (MLLM)
$\mathbf{P}_{\text{gt}},\ L_{\text{gt}}$
GT Solution
I_{input}
Input Image
Text Encoder
f_ϕ
h_{ref}
Reference State
$\mathcal{L}_{\text{align}}$
Final Thought
Condition Signal
Diffusion Transformer
f_ψ
$z_\tau(1) \sim \mathcal{N}(0, I)$
$z_\tau(t)$
$I_\tau = z_\tau(0)$
Noise
Diffusion Process $t \in [0, 1]$
Target Image
Conditional Generation (DiT)